

## Section 04



# Industrial Security (Bachelor)

Section 04 – Industrial Protocols

## Industrial Protocols

*Lecture Notes*

Department of Computer Science

- 1 Legacy Heritage
- 2 Modbus
- 3 Siemens, DNP3, EtherNet/IP
- 4 Modern Secure Protocols

## By the end of this section you will be able to

- Recognise common industrial protocols and their use cases.
- Understand why legacy protocols typically lack authentication.
- Decode a Modbus TCP frame by hand.
- Compare insecure legacy protocols with secure modern alternatives (OPC UA, MQTT-TLS).
- Identify when secure variants are appropriate.

# 1

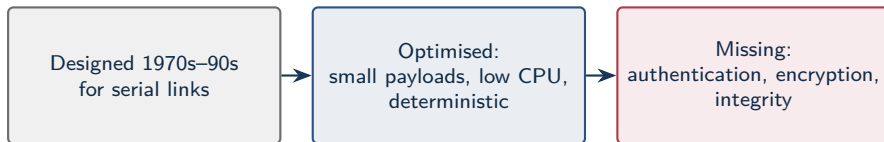
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## Legacy Heritage

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## Core fact

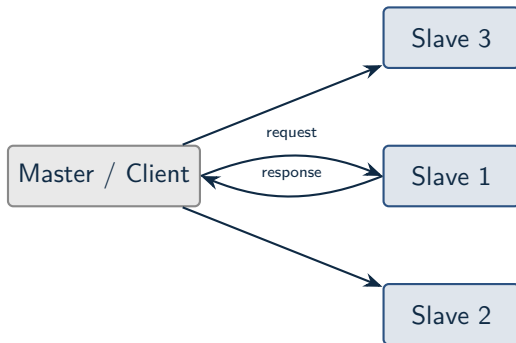
Most industrial protocols in production today were **designed before the Internet** and provide no authentication, integrity, or confidentiality.



# 2

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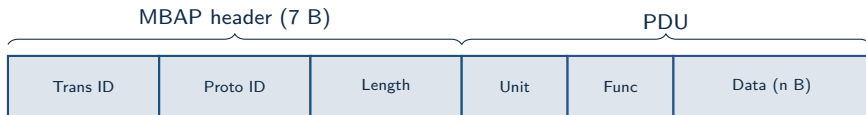
## Modbus



Variants: Modbus RTU (serial), Modbus TCP (port 502)

Address space: coils, discrete inputs, input regs, holding regs

No auth, no encryption – by design (1979)



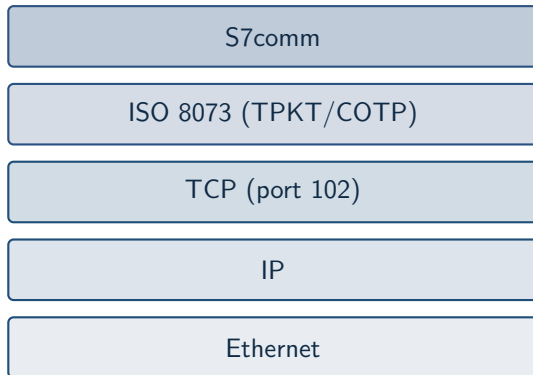
```
1  00 01 00 00 00 06 01 03 00 00 00 0A
```

```
2  ^TID^ ^PID^ ^LEN^ ^U^ ^FC^ ^Addr^ ^Cnt^    -> Read 10 Holding Regs from 0x0000
```

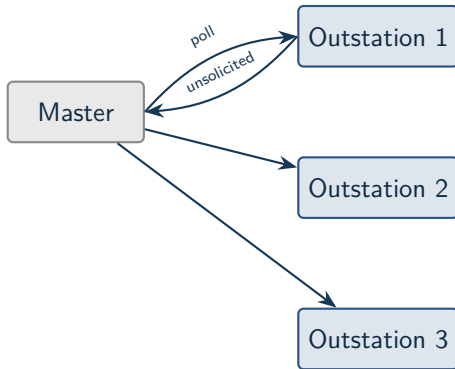
# 3

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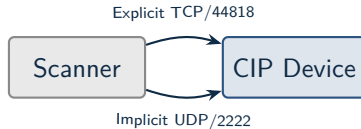
## Siemens, DNP3, EtherNet/IP



Read/write variables, upload/download programs, start/stop CPU. S7commPlus added session keys – still broken in practice.



DNP3 dominant in NA utilities; IEC 60870-5-104 in Europe. Both ride TCP today; both added Secure Authentication (SAv5) on top, but encryption is optional.



CIP carries object-oriented messages (assemblies, parameters, identity).  
CIP Security adds TLS/DTLS and certificate-based authentication.



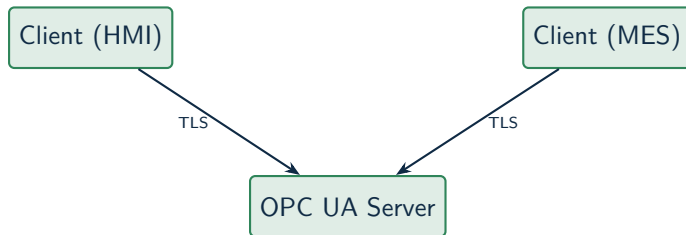
EtherType 0x8892 – bypasses IP for real time.  
 IRT needs hardware switching for sub-ms cycles.

# 4

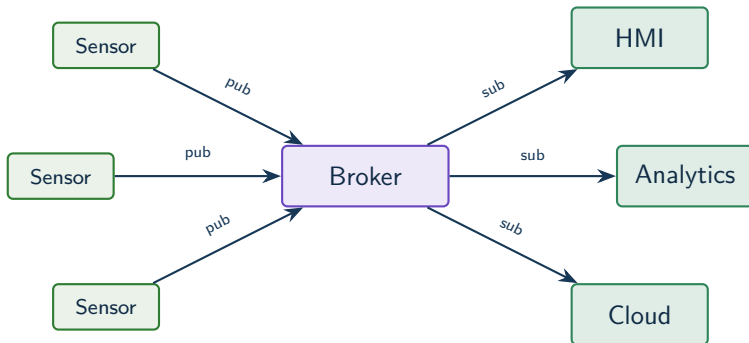
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## Modern Secure Protocols

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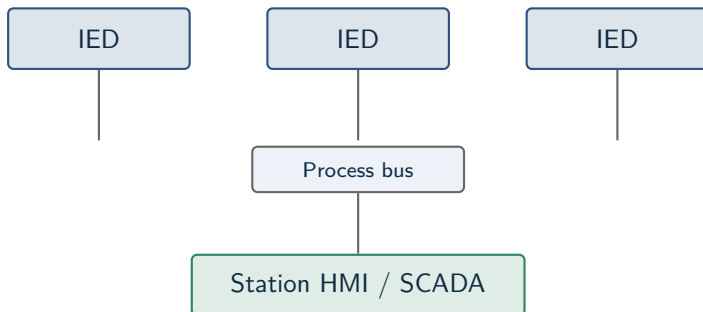


X.509 mutual auth • message signing • encryption  
Rich information model • pub/sub since v1.04



Default 1883 plaintext / 8883 with TLS. Sparkplug B adds OT semantics.

| Protocol    | Auth     | Integrity | Confid. | Notes               |
|-------------|----------|-----------|---------|---------------------|
| Modbus      | ×        | ×         | ×       | 1979, ubiquitous    |
| S7comm      | ×        | ×         | ×       | Siemens-proprietary |
| DNP3        | opt SAv5 | opt       | ×       | Utilities           |
| EtherNet/IP | opt CIP  | opt       | opt     | Rockwell            |
| Profinet    | weak     | ×         | ×       | Siemens / IRT       |
| OPC UA      | X.509    | sign      | TLS     | modern default      |
| MQTT        | +TLS     | TLS       | TLS     | IIoT messaging      |



## The substation stack

**MMS** for client/server; **GOOSE** for sub-4 ms trip messages; **Sampled Values** for measurements. Goose has *no* authentication by default.

### ★ Key Takeaway: What to remember from this section

- Most legacy industrial protocols are unauthenticated and unencrypted by design.
- Modbus is the canonical example – 12 bytes of MBAP and a one-byte function code.
- Modern protocols (OPC UA, MQTT-TLS) bring real cryptography, but require correct configuration.
- Network-layer mitigations (firewalls, segmentation) compensate where the protocol cannot.

# End of Section 04



Industrial Protocols

Questions and discussion